

Revised Sections for Referee Submission

Section 2 (Taxonomy Definition) · Research Directions

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April 2026 · Zenodo DOI: 10.5281/zenodo.19117400

These two sections replace the original Section 2 ("Derivation of the Stability Threshold $g = 33$ ") and the implicit retrocausality claims in Step 4 of the proof. All other sections of GTCT-2026-001 remain unchanged. Cross-references are updated below.

2. The g-Series: A Regime Taxonomy

The dm^3 -GTCT framework uses a taxonomy of behavioral scales called the **g-series**. This section defines the taxonomy precisely. The g-series makes no mathematical claims about the index values; the indices are labels, chosen by the criteria stated below, not derived constants.

Definition 2.1 (g-Series Taxonomy).

The g-series is a finite ordered set of regime labels

$$g^0 < g^2 < g^6 < g^{33} < g^{64}$$

Each label g^k names a qualitative behavioral regime of a dm^3 system. The superscript k is an ordinal index, not an exponent. The series is not a sequence of numbers; it is a classification.

Definition 2.2 (Index Criteria).

The five indices are assigned by the following criteria, each corresponding to a qualitatively distinct stability property of the operator chain $G = U \blacksquare F \blacksquare K \blacksquare C$:

Label	Index	Regime Name	Criterion
$g^{\blacksquare}$	0	Seed	Single operator application; no closure condition.
g^2	2	Compositional	Two-operator closure; minimal self-reference (F active).
$g^{\blacksquare}$	6	Cycle-scale	Full G-cycle closure; limit cycle Gamma entered.
g^{33}	33	Soft equilibrium	Heuristic stability index: $\text{ceil}(\log_2(3!) * 4) * 3 = 33$. See Remark 2.3.
$g^{\blacksquare\blacksquare}$	64	Phase threshold	Complete possibility-space coverage; $2^6 = 64$ operator-state combinations.

Remark 2.3 (Status of the index 33).

The index 33 arises from the following heuristic computation: there are three invariants that must close simultaneously (orthogonality, nilpotency of deviations, spectral collapse), giving $3! = 6$ orderings; $\log_2(6) \cdot 4 \approx 10.34$, so $\text{ceil}(\log_2(3!)) \cdot 4 = 11$; multiplying by 3 for a triple-confirmation robustness criterion gives $3 \times 11 = 33$. *This computation is heuristic. The "multiply by 3" step reflects a design choice — that three independent closures are required for robust stability — which is not derived from the contact-geometric structure of Sections 5–6. The index 33 is therefore a motivated label, not a theorem. Its mathematical status is that of a conjecture: we conjecture that dm^3 systems require at least 33 operator cycles for the three invariants to achieve simultaneous robust closure, and we identify this as an open verification target for the AXLE Lean 4 repository.*

Remark 2.4 (Crystal notation).

The notation $g^6:33$ and $g^{33}:33$ used elsewhere in the Principia Orthogona series denotes association, not equality. " $g^6:33$ " means: the cycle-scale regime g^6 carries the structural index 33. It is a descriptor of the same kind as " D_6 symmetry: 6-fold" or " E_8 lattice: rank 8". No algebraic identity $g^6 = 33$ is claimed.

Cross-reference update. All references in Sections 3–6 to "the derivation of $g = 33$ " should be read as references to Definition 2.2 and Remark 2.3 above. The stability radius $\varepsilon_0 = 1/3$ (Section 5) is independent of the g -series taxonomy and remains a theorem.

8. Research Directions

This section replaces the retrocausality claims in Step 4 of the original proof. The contact-geometric results of Sections 5–6 are classical dissipative dynamics; they do not imply retrocausality. The connections described below are structural analogies and open research problems, not theorems.

8.1 Operator Grammar and Time-Symmetric Quantum Mechanics

The two-state vector formalism (TSVF) of Aharonov and Vaidman [4] assigns both a forward-evolving and a backward-evolving state to each quantum system. The operator chain $G = U \blacksquare F \blacksquare K \blacksquare C$ has a structural resemblance to this bidirectional picture: the Fold operator F introduces self-reference at mid-circuit, and the Unfold operator U projects outward. Whether this resemblance can be made precise — by constructing a map from the dm^3 contact system to the TSVF Hilbert-space formalism — is an open problem. No retrocausal claim is made in this paper.

Open problem 8.1. Construct, if possible, a functor from the dm^3 operator category to the category of pre- and post-selected quantum states in the TSVF, preserving the stability radius $\varepsilon_0 = 1/3$.

8.2 Delayed-Choice Experiments and the Fold Operator

In the delayed-choice quantum eraser (Kim et al. [5]), a future measurement choice determines which past path information is available. The Fold operator F — which introduces the first self-reference in the chain — is structurally analogous to the beam-splitter choice in that experiment: it is the point at which the system curves back onto its own prior state. This analogy motivates the following research direction, but does not constitute a proof of any physical connection.

Open problem 8.2. Identify an experimental observable in a quantum optics or NMR system that would distinguish "spiral enrichment" ($x' \neq x$ after circuit closure) from linear return ($x' = x$), and determine whether the dm^3 stability threshold g^{33} has a measurable counterpart in that system.

8.3 Entanglement Swapping and the Spiral Return

Ma et al. [6] demonstrated entanglement swapping between photons that never coexisted, suggesting non-trivial temporal structure at the quantum scale. The spiral return $x \rightarrow x'' \rightarrow x'$ with $x' \neq x$ in the dm^3 circuit is formally analogous: the source state is enriched by one completed circuit without returning to its original configuration. Whether this formal analogy corresponds to a physical mechanism is unknown.

Open problem 8.3. Determine whether the contact-geometric spiral return, formalized via the Floquet multiplier $\lambda_{\text{transverse}} = e^{-4\pi i}$, admits a quantum information interpretation consistent with the entanglement swapping results of Ma et al.

8.4 Negative Group Delay and the Weave Scale

Angulo et al. [7] report experimental evidence of negative group delay — photons appearing to exit a medium before entering. This is a well-understood phenomenon in classical wave optics (anomalous dispersion) and does not require retrocausality. The dm^3 framework does not explain or predict negative group delay. The reference is included as context for the broader research landscape, not as evidence for the operator grammar.

8.5 Formal Verification in AXLE (Lean 4)

The AXLE repository (github.com/TOTOGT/AXLE) contains Lean 4 proof obligations for the dm^3 framework. The highest-priority open items relevant to this paper are:

- (a) Formal verification of the Gronwall bound and $\varepsilon_0 = 1/3$ (Section 5) — currently the most tractable target.
- (b) Formalization of the g-series taxonomy (Definition 2.2) as a Lean 4 inductive type, with the index-33 conjecture stated as a sorry to be discharged by future work.
- (c) The Poincare-to-Collatz correspondence — the hardest open item, currently marked as a sorry in AXLE — is not required for the results of this paper.

Note on References

References [4]–[8] (Aharonov-Vaidman, Kim et al., Ma et al., Angulo et al., Herczeg-Waldron) are retained in the bibliography as motivation for the research directions in Section 8. None of these references is cited as support for a theorem in this paper. The causal and retrocausal claims in Step 4 of the original proof are withdrawn and replaced by Open Problems 8.1–8.3 above.